Case Study Portfolio: Project Velopair

2026

Velopair

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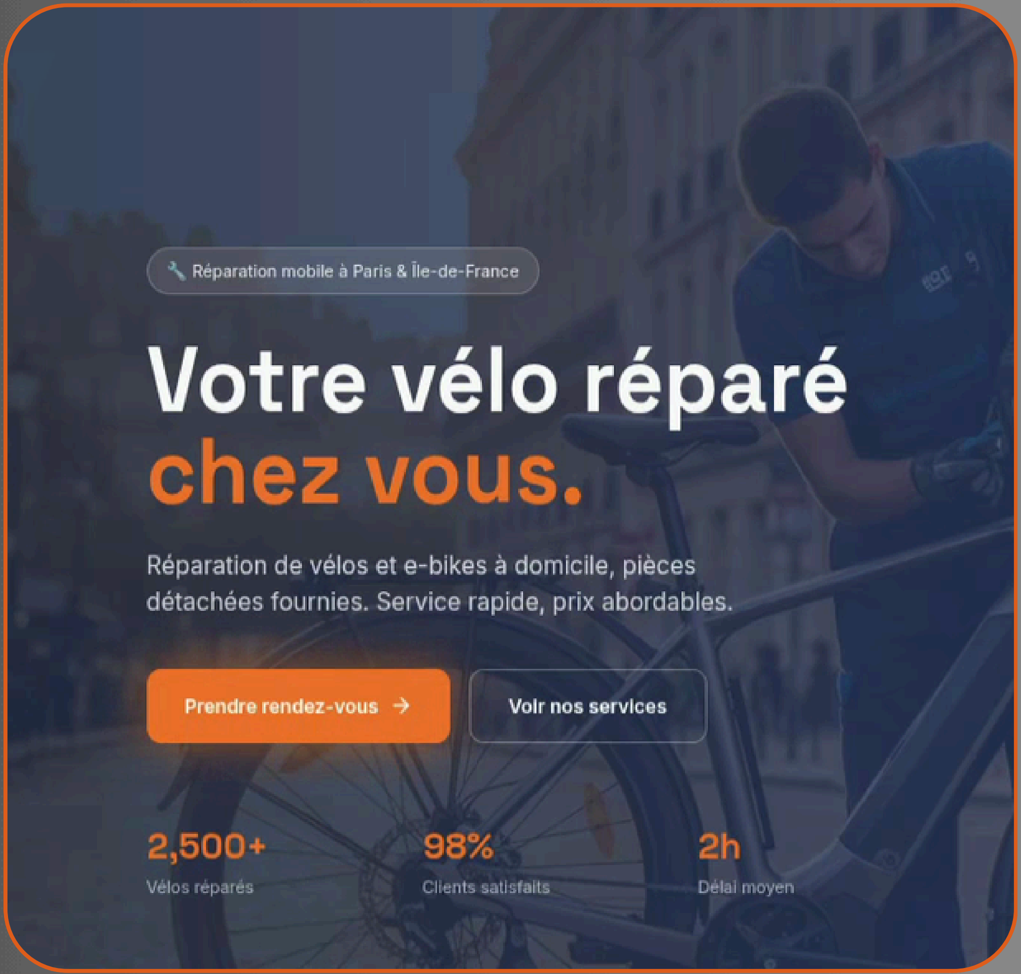
Restoring Hidden Ad Conversions

Velopair's ad campaigns were operating with massive blind spots. Traditional web tracking hooks are crumbling under modern privacy settings.

This project details how I migrated their data pipeline off the unstable user browser and onto a private cloud server—reclaiming lost attribution data while remaining completely compliant with privacy laws.

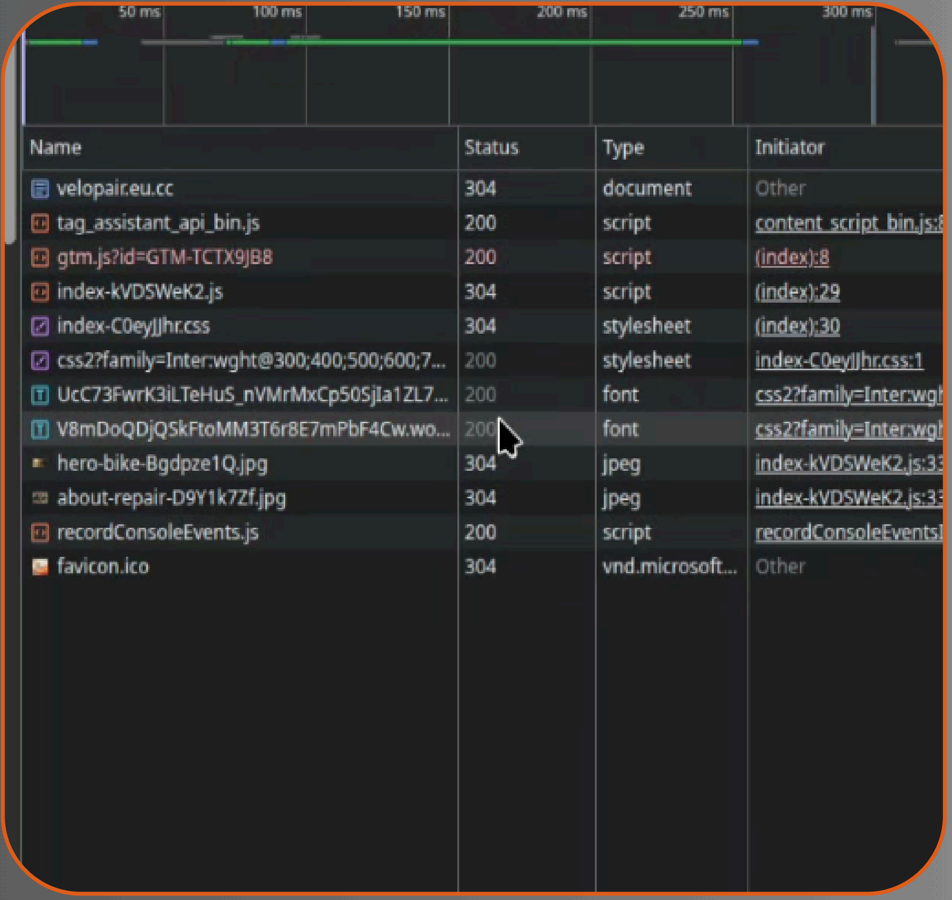


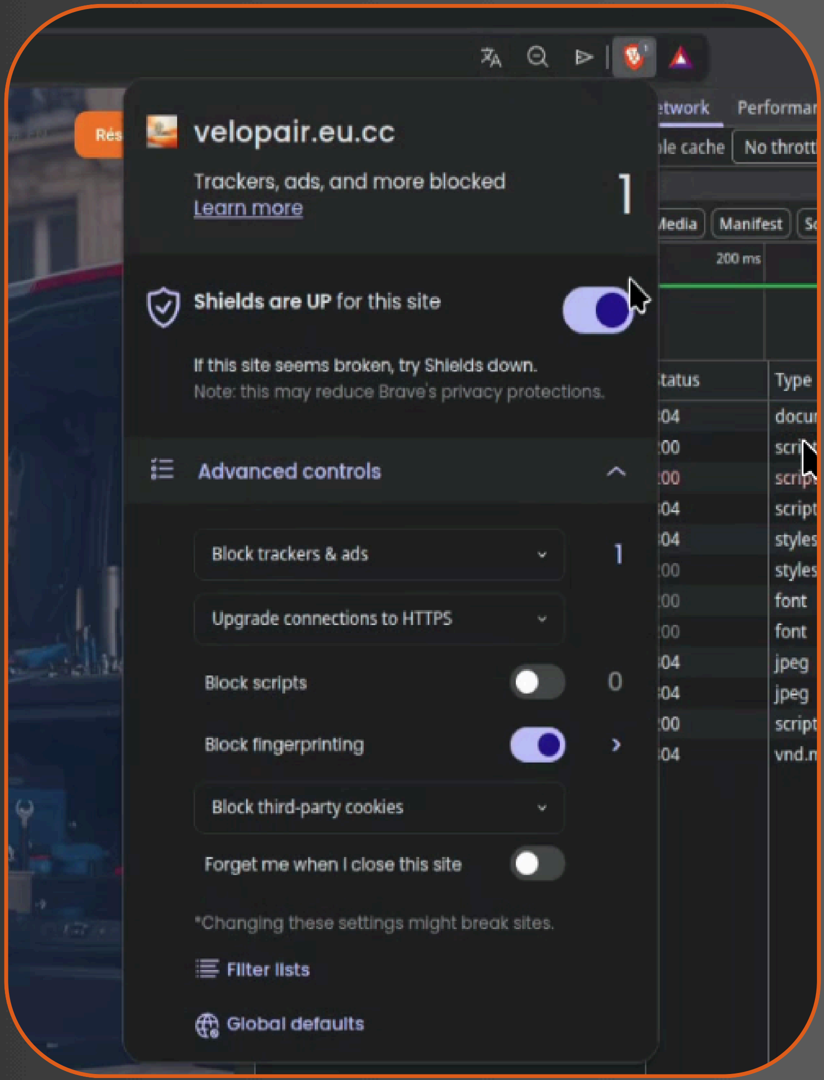
The Fatal Flaw of Client-Side Tracking

Most e-commerce stores still use standard "client-side" tracking, meaning third-party marketing scripts run directly inside the visitor's browser.

The moment a user tightens their browser security or turns on an ad blocker, those scripts are instantly killed.

For a business, this means missing conversion events and severely degraded ad performance.



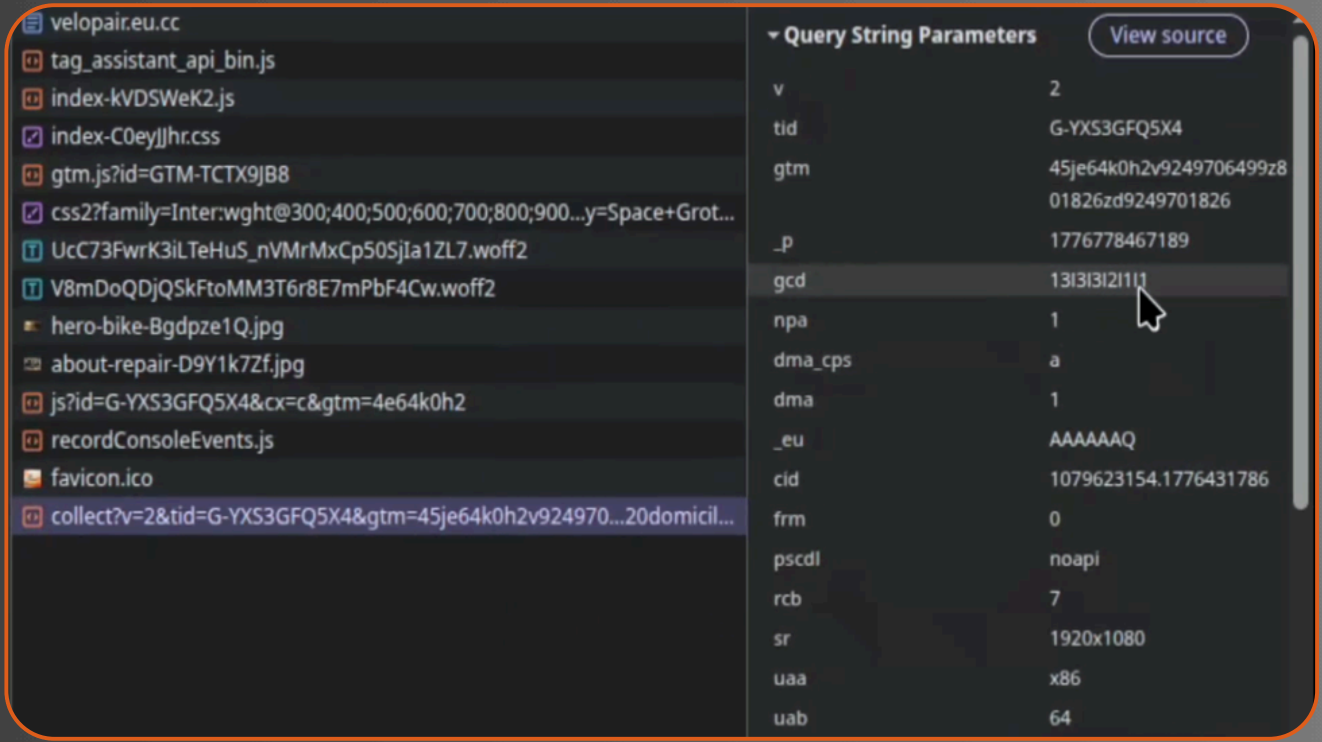


Total Signal Failure Under Strict Browsers

To demonstrate the problem, I tested the baseline setup inside a hardened browser environment like Brave with ads shields on.

The browser's built-in blocklists target signature files like gtm.js instantly.

The tracking framework is completely paralyzed before a single click, page_view, generate_lead event can even be registered.



Messy Consent Management = Lost Data

Tracking isn't just about collecting data;
it's about legality. When cookie banners are unstable or poorly integrated with the data layer, variables frequently reset to undefined.

Tags try to fire out of order, creating massive compliance risks and leaving data analysts with fragmented, unusable user sessions.

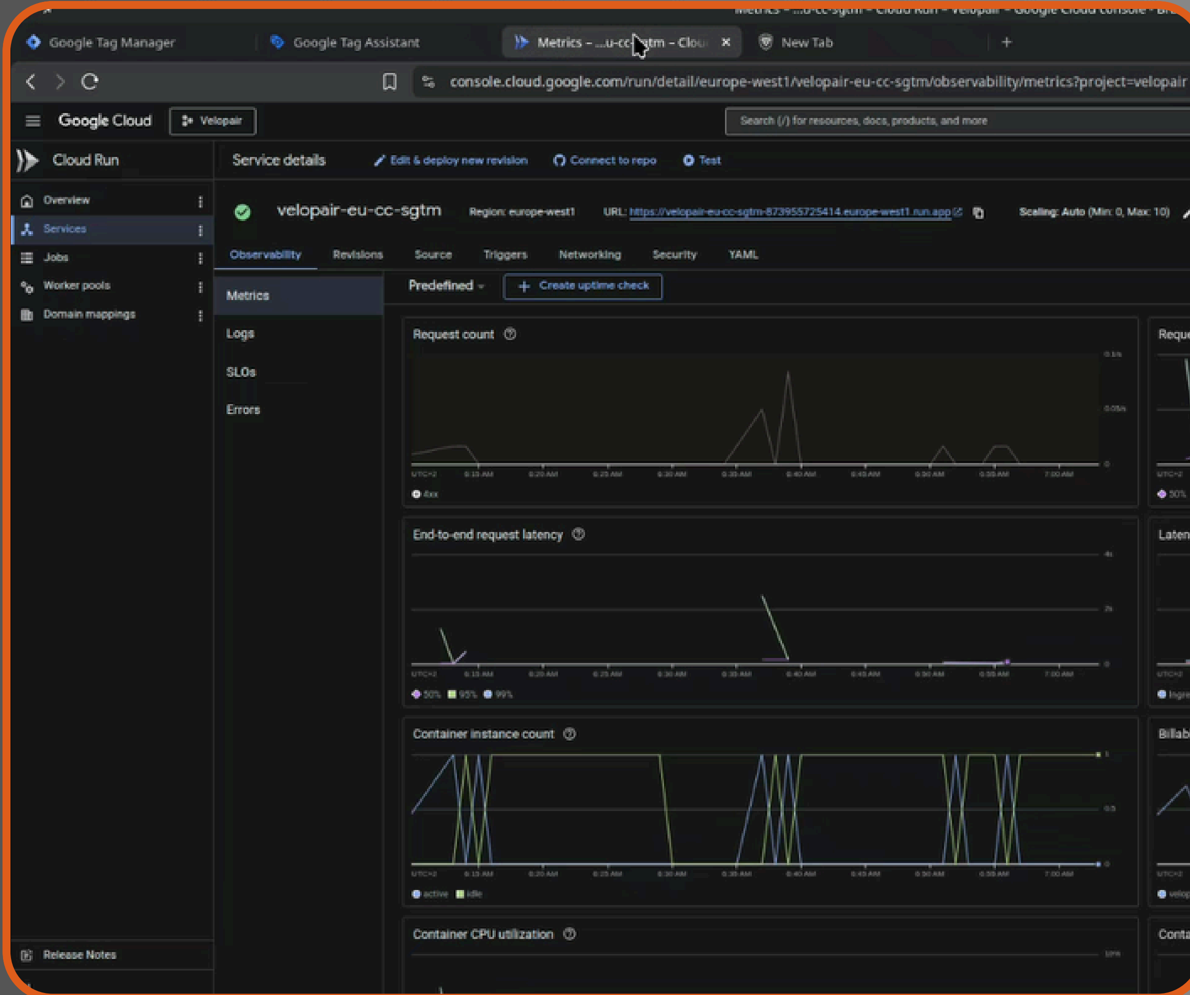
Moving the pipeline to Google Cloud Run

The solution was to take the tracking load out of the browser entirely.

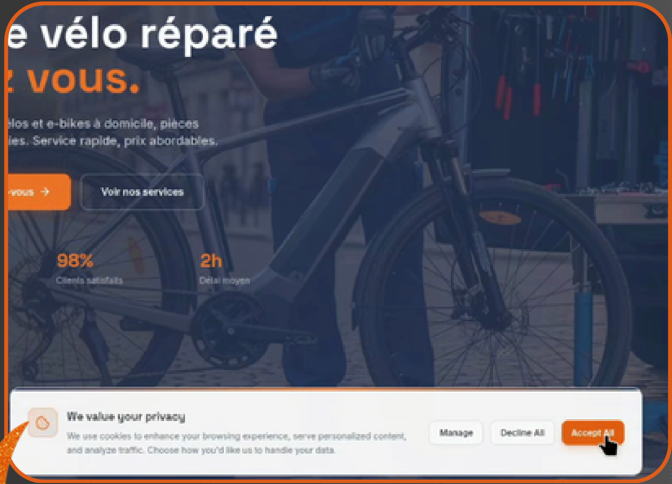
I engineered a private server-side tracking gateway deployed on serverless architecture inside Google Cloud Platform (GCP).

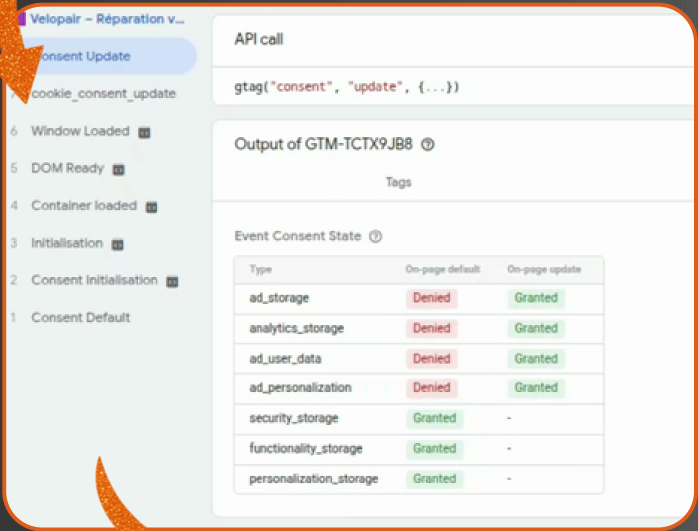
Now, the website routes all event data through a secure, first-party subdomain.

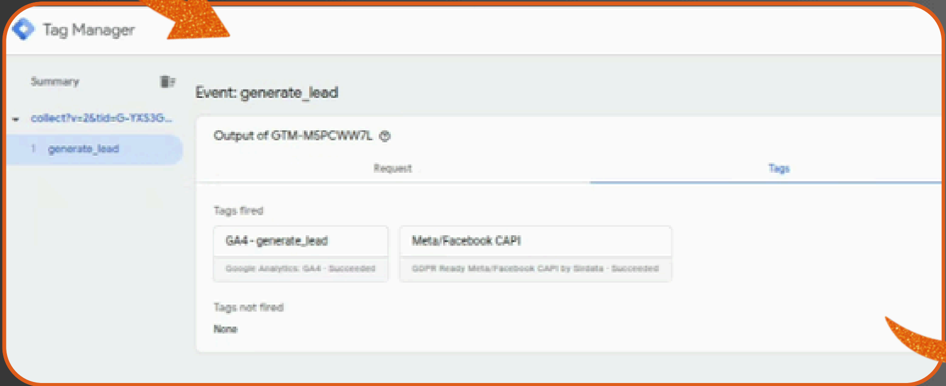
Because the data stream looks like standard website traffic, browser-level content blockers cannot disrupt it.

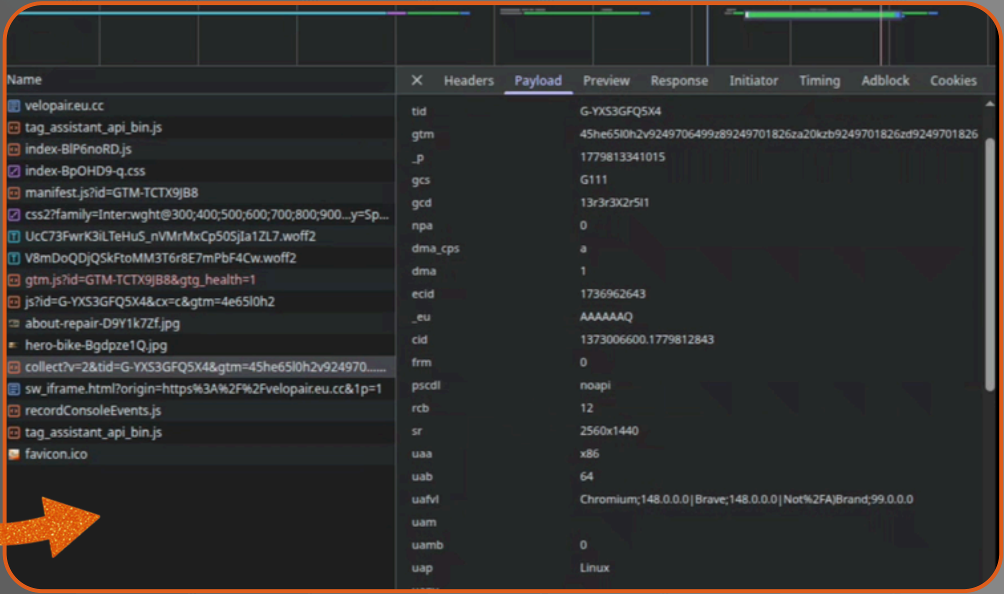


Integrating Google Consent Mode V2









I didn't just bypass adblockers; I did my best to respect user privacy and compliance programmatically.
I configured Google Consent Mode V2 directly within the server container.

The system reads the user's choice instantly.
If a user rejects cookies, the server automatically strips away all identifiable data but still relays a secure, anonymous signal to Google so ad algorithms can optimize without violating GDPR / RGPD rules.

Real-Time SHA-256 Hashing for Meta CAPI

Output of GTM-M5PCWW7L

Request | Tags

Variable	Variable type	Return type	Value
_event	Custom Event	string	"generate_lead"
CAPItoken_expires_13.07.2026	Constant	string	"EAAMH9r1cg30BRTfOz1kAeLyxbLZAaessHkC1UxA6ecQ6omBazHJB8REaWjAmRIKjbjT" + "x88gC8tcmTV98iRXyQZBiE2ZBZAhmGZBRguSFrimHSQzjOwl6wKdXIkVldCKBDA6AicD" + "CgSxrVsOMyJpFkhAh3NXyASYA2t7vZAhtYKfghm9ftdxh2McZB5bZBoDCNW72"
Client Name	Client name	string	"GA4"
Container ID	Container ID	string	"GTM-M5PCWW7L"
EDV - Email Hashed	Phone/Mobile Normalizer & SI	string	"5ac4876b204d837b3352a2ceb95f3024de4faeef65ad4bf68c2aeed187a9f94e"
EDV - Email Raw	Event Data	string	"josephtasdigital@outlook.com"
EDV - Event Name	Event Data	string	"generate_lead"
EDV - fbp	Event Data	undefined	undefined
EDV - Message - Hashed	Phone/Mobile Normalizer & SI	string	"9feaabed2d1fb050cd6d9f95e5aa9811935129d4d83c2f609057a7a845cf77ef"
EDV - Message - raw	Event Data	string	"no probs boss"
EDV - Name - Raw	Event Data	string	"joseph"
EDV - Name Hashed	Phone/Mobile Normalizer & SI	string	"6e1cbc9c5fd02390de4ac6a231625752651 88b7dff2934b723eba1ff3ff43d5a"
EDV - Phone Hashed	Phone/Mobile Normalizer & SI	string	"abfc258b27b0b90fad4775f823d0a752cadd5caef82e53330ab2c6ab4be2af0b"
EDV - Phone Raw	Event Data	string	"+123456789"
EDV- fbc	Event Data	undefined	undefined
Event Name	Custom Event	string	"generate_lead"
Meta Capi test code	Constant	string	"TEST45632"
Meta Pixel ID	Constant	string	"4358126601104442"
Visitor Region	Visitor Region	string	"FR-NOR"
Web gtm container	Constant	string	"GTM-TCTX9JB8"

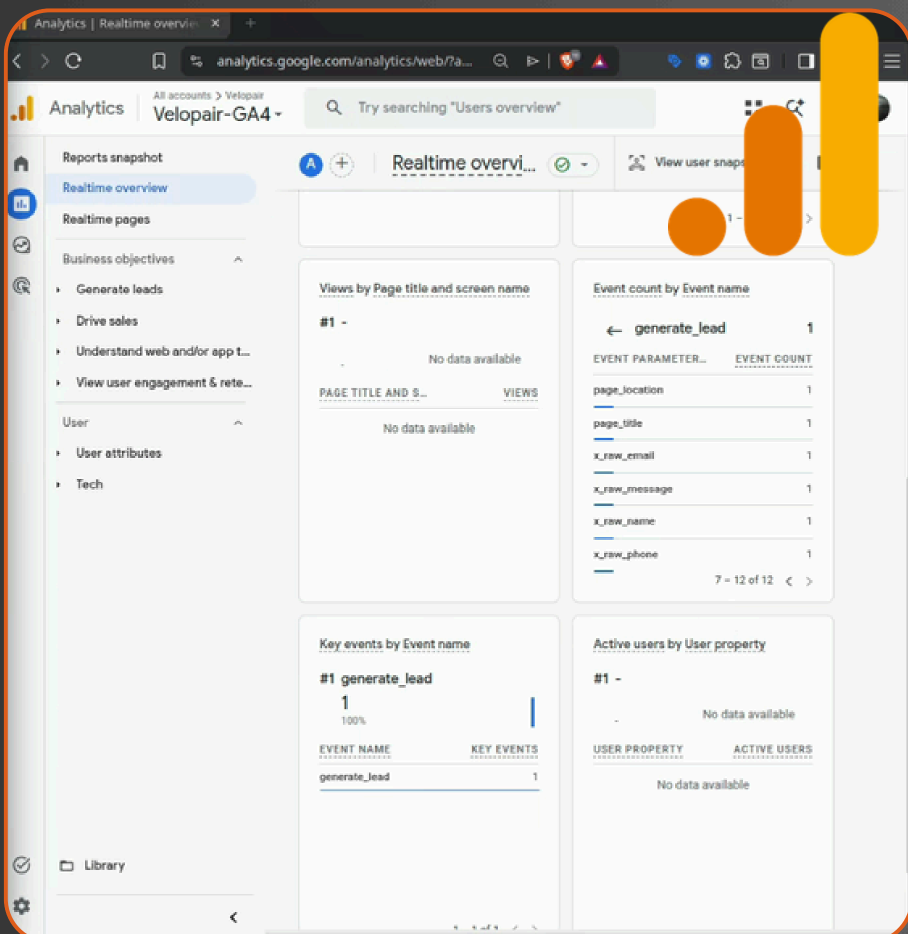
To maximize Meta Conversions API (CAPI) match quality, you must send user signals like emails or phone numbers.

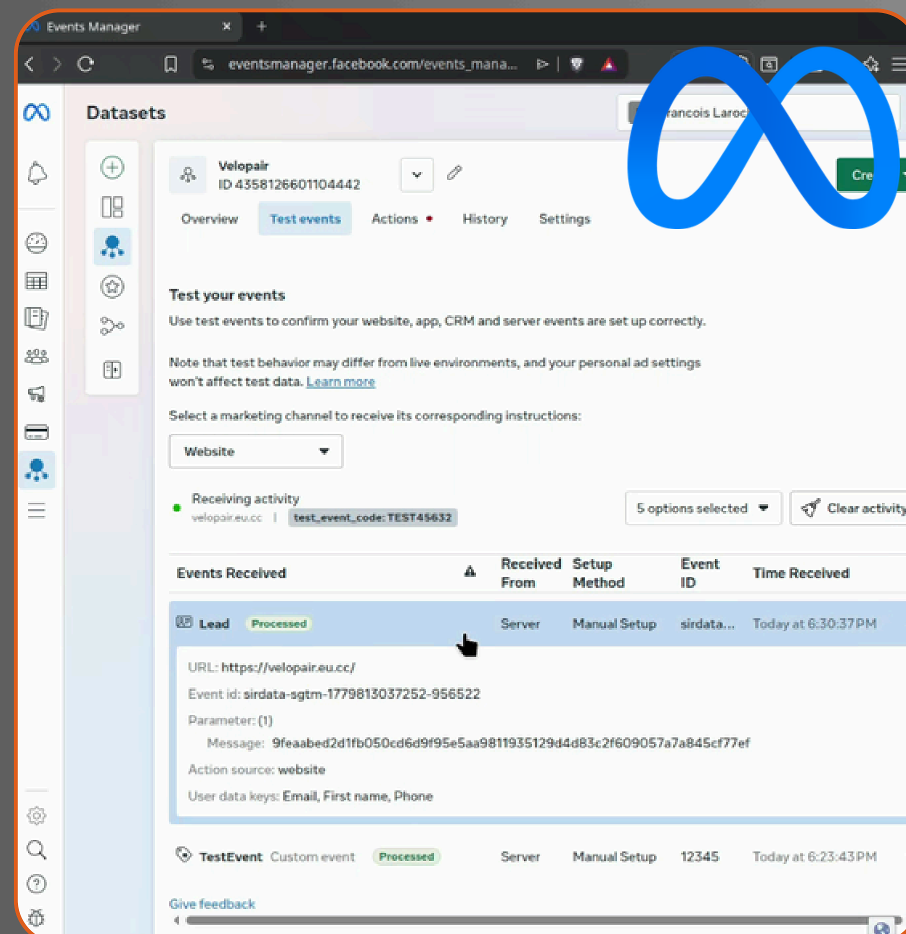
However, routing raw customer profiles across the web is a massive data liability.

I added a custom hasher within the server container to intercept incoming data layers and instantly encrypt sensitive inputs using cryptographic SHA-256 hashing before forwarding them to ad networks.

Complete Signal Recovery Proven

The ultimate test:
submitting a conversion form with all aggressive browser blocks fully engaged.
The result was flawless.
The server-side pipeline intercepted the event smoothly, bypassing the browser's wall entirely.
The conversion landed simultaneously inside GA4's DebugView and Meta's Events Manager with matching deduplication IDs.





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